

HNSS04 · Initial Formation of Craniofacial Derivatives — Overview

Everything on one page: when each structure forms, where it comes from, and what goes wrong. Detail charts follow for the tongue, thyroid and face.

	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9	Wk 10	Later (wk 11 → puberty)
Tongue		Wk 4–5 Arch 1: 2 lateral lingual swellings + tuberculum impar	Wk 5–10 · Copula (hypobranchial eminence, arches 2–4) → posterior 1/3 · lateral swellings overgrow the tuberculum impar → anterior 2/3 · occipital somite myoblasts migrate in → tongue muscle					
Thyroid		Wk 4–5 Endoderm buds at foramen cecum; descends as a bilobed diverticulum on the thyroglossal duct	Wk 6 Front of hyoid + laryngeal cartilages	Wk 7 Final spot: front of the trachea			Wk 14–15 Starts working: T4/T3 (colloid) + calcitonin	
Face surface		Wk 4–5 5 prominences + nasal placodes	Wk 5–8 · Placodes → nasal pits → medial + lateral nasal prominences · maxillary prominences grow medially and fuse → upper lip, nose		Wk 8–9 Cheeks identifiable			
Palate & nasal cavity			Wk 5–8 · Intermaxillary segment → primary palate	Wk 6.5–7 · Palatine shelves grow over the tongue → secondary palate				
			Wk 6 → mid-wk 7 · Oronasal membrane ruptures → choana				→ Puberty · Paranasal sinuses keep growing	
Teeth not tested			Wk 6 Dental lamina (C-shaped) → 10 buds/jaw → cap stage		Wk 8–15 · Milk teeth first; secondary (permanent) buds begin forming			

Where each structure comes from

STRUCTURE	ORIGIN
Anterior 2/3 of tongue	Arch 1: lateral lingual swellings + tuberculum impar
Posterior 1/3 of tongue	Copula / hypobranchial eminence (arches 2–4)
Tongue muscles	Occipital somites (mesoderm) → CN XII
Thyroid follicles	Endoderm at the foramen cecum
Thyroid C cells	Ultimobranchial body (pouch 5/4)
Upper lip	2 medial nasal + 2 maxillary prominences (not lateral nasal)
Nose	5 prominences: frontonasal (bridge), medial nasal (crest, tip), lateral nasal (sides)
Primary / secondary palate	Intermaxillary segment / palatine shelves of maxillary prominences
Facial bones	Neural crest (exceptions: thyroid + cricoid cartilages; ? malleus/incus)

Skull: neurocranium vs viscerocranium

	NEUROCRANIUM	VISCEROCRANIUM
What	Surrounds the brain	Face & jaws (from the arches)
Ossification	Mostly endochondral (cartilage model first)	Mostly membranous (bone forms directly)
Source	Paraxial + prechordal mesoderm + NC	Neural crest + some mesoderm

>100 ossification centers → 45 bones → fuse into the 22 bones of the adult skull. Each part has a cartilaginous and a membranous subdivision.

Clinical problems

- Thyroglossal duct cyst/fistula: midline, anywhere along the descent path (below hyoid, base of tongue)
- Cleft lip: maxillary + medial nasal fail to fuse (lateral cleft lip ~1/1,000, 80% male)
- Cleft palate: palatine shelves fail to fuse · oblique facial cleft: maxillary + lateral nasal fail
- Treacher Collins: Tcof-1 → NC migration from R2/R4 fails
- Down syndrome on ultrasound: absent nasal bone, abnormal ear position

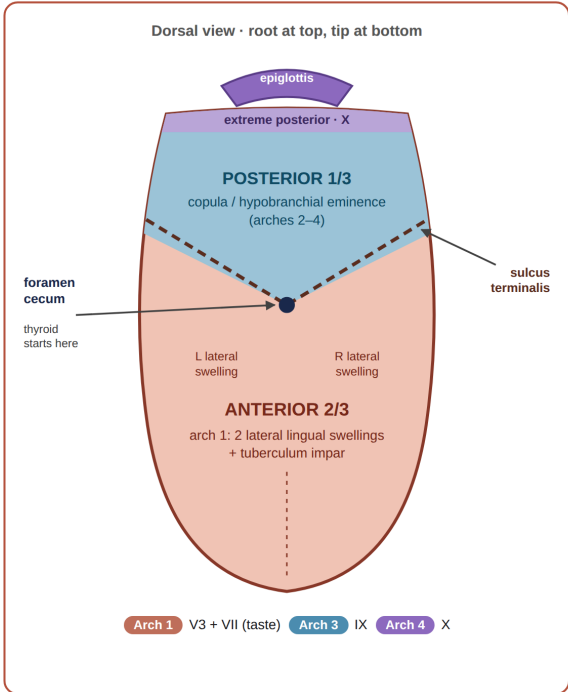
Recurring theme (tongue · lung · thyroid):

the underlying mesoderm + neural crest signals the overlying endoderm to proliferate and bud. The thyroid forms “much like the lungs” but buds only twice (bilobed).

★ USMLE high-yield: Tongue muscles = occipital somites → CN XII (palatoglossus = X). Thyroid starts at the foramen cecum; most common ectopic site = lingual thyroid. Cleft lip = medial nasal + maxillary fusion failure; cleft palate = palatine shelf fusion failure (separate causes, often together).

HNSS04 · The Tongue

Each region of the tongue keeps the nerve of the arch it came from. The muscles are the exception: they migrate in from the occipital somites.



How the tongue forms

WHEN	STRUCTURE	FATE
Wk 4–5	Arch 1: two lateral lingual swellings + one medial tuberculum impar	Lateral swellings overgrow the tuberculum impar and merge → anterior 2/3
Wk 5–10	Second medial swelling: the copula (anterior hypobranchial eminence), from mesoderm under arches 2, 3, 4	Posterior 1/3 (root)
Wk 5–10	Myoblasts migrate in from the occipital somites (paraxial mesoderm)	Tongue muscles
Wk 5–10	Posterior hypobranchial eminence (arch 4) + epiglottal swellings	Epiglottis + primordial glottis; the solid glottis covering opens by apoptosis → laryngotracheal orifice

Tissue rule: the tongue is mesoderm surrounded by endoderm. The foramen cecum sits between the tuberculum impar and the hypobranchial eminence; it's where the thyroid starts.

Innervation: region follows its arch

REGION	GENERAL SENSATION	TASTE	ARCH
Anterior 2/3	V3 (lingual n.)	VII (chorda tympani)	1
Posterior 1/3	IX	IX	3 (copula)
Extreme posterior	X (superior laryngeal)	X	4
Muscles	XII (hypoglossal) — palatoglossus = X		Occipital somites

Throat (pharynx) muscles by nerve

Constrictors, levator palatini	X (arch 4)
Stylopharyngeus	IX (arch 3)
Tensor palatini	V3 (arch 1)
Cricothyroid / other larynx	X: superior laryngeal / recurrent laryngeal
Gag reflex	Afferent IX · efferent X

Tongue ↔ lung: the shared mechanism

Mesoderm + neural crest → signals → overlying endoderm → endoderm proliferates

→ tongue swellings / lung buds

Same epithelial–mesenchymal theme builds the lung (and the thyroid, which buds “much like the lungs” but only twice).

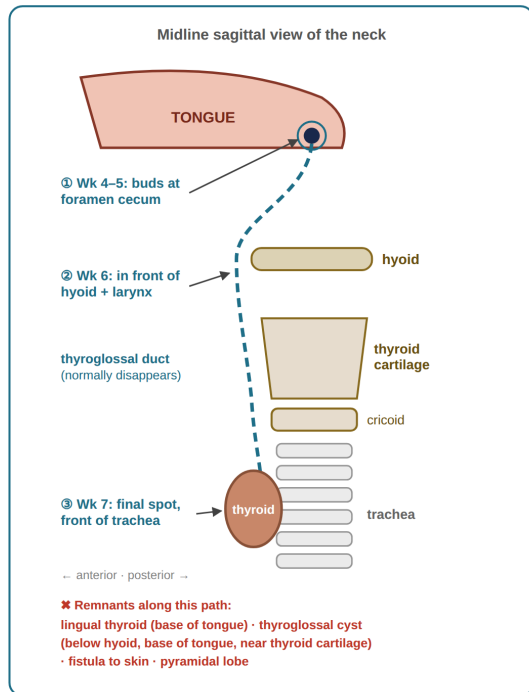
Molecular detail (slides marked not on exam): endoderm BMP-7 blocks growth → mesoderm Noggin inhibits BMP-7 → Patched released → endoderm proliferates. Hox proteins carried in by NC drive the mesenchyme.

Hook: “Front is 1 (V3 + VII taste), back is 3 (IX), the very back is 4 (X) — and the muscles came from the neck (occipital somites → XII).”

★ USMLE high-yield: CN XII lesion → tongue deviates toward the lesion. Bell's palsy (VII) → lost taste on the anterior 2/3, but touch intact (V3). IX lesion → lost taste + sensation on the posterior 1/3 and lost gag (afferent). Palatoglossus is the one tongue muscle on X.

HNSS04 · The Thyroid Gland

Born at the back of the tongue, it walks down the front of the neck on a duct that should vanish. Anything left behind along that path is a clinical problem.



Steps of thyroid development

- 1 Endodermal epithelium proliferates at the base of the pharynx at the **foramen cecum**, between the **tuberculum impar** and the **hypobranchial eminence**. Forms “much like the lungs.”
- 2 The primordium **descends** from the foramen cecum as a **bilobed diverticulum** (buds only twice, unlike the lung). **BMPs** are expressed at the **tip** of the descending gland.
- 3 It stays attached to the tongue by the **thyroglossal duct**, which later **disappears by apoptosis**.
- 4 **Week 6:** lies in front of the **hyoid bone** and **laryngeal cartilages**.
- 5 **Week 7:** reaches its **final position** in front of the **trachea**.
- 6 The **ultimobranchial body** (pouch 5/4) joins and supplies the **C cells**.

PAX marks **neural crest** → PAX expression in the thyroid means **NC contributes to the thyroid gland** **EXAM EMPHASIS**
Other genes: TTF-1 (thyroid transcription factor), BMPs, noggin

Fetal function — starts at weeks 14–15

Follicular cells

Produce **colloid**, the source of **thyroxine (T4)** and **triiodothyronine (T3)** → control **cellular metabolism**.
Levothyroxine = synthetic T4.

C cells (ultimobranchial body)

Secrete **calcitonin**, a 32-amino-acid peptide → binds receptors on osteocytes/osteoclasts → ↑ Ca^{2+} uptake from fetal blood → **prevents bone resorption** (net bone gain). Needed for **normal bone development**.
Fetal levels are 2–3× maternal.

Clinical problems

Thyroglossal duct cyst	Cystic remnant of the duct; midline , anywhere along the path (below the hyoid , base of tongue, near thyroid cartilage). Moves up with swallowing / tongue protrusion. Generally not a major clinical problem to correct.
Thyroglossal fistula	Duct remnant opens to the skin by a canal; can be present at birth
Lingual thyroid	Thyroid fails to descend → mass at the base of the tongue (most common ectopic site)
Pyramidal lobe	Persistent lower end of the duct (normal variant)

Don't confuse

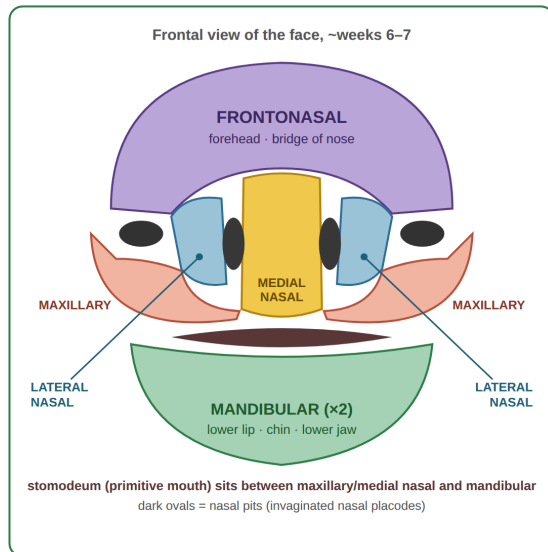
	THYROGLOSSAL DUCT CYST	BRANCHIAL CLEFT CYST
Position	Midline	Lateral , anterior to SCM
Swallowing	Moves	Does not move
Origin	Thyroid descent path	Persistent cervical sinus (cleft 2)
Lining (HNSS16)	Respiratory epithelium + thyroid follicles	Squamous + lymphoid tissue

Hook: “Born in the tongue, walks down the neck.” Foramen cecum (wk 4–5) → front of hyoid/larynx (wk 6) → front of trachea (wk 7) → switches on (wk 14–15).

★ **USMLE high-yield:** Midline neck mass that **moves with swallowing** = thyroglossal duct cyst. Before removing a **lingual thyroid**, confirm other functioning thyroid tissue exists — it may be the only thyroid. **Congenital hypothyroidism (cretinism):** pot belly, pale puffy face, protruding umbilicus, large tongue, impaired brain development — caught by newborn screening; early levothyroxine prevents intellectual disability.

HNSS04 · The Face (Lips, Nose, Cheeks, Palate)

Five prominences surround the primitive mouth (stomodeum). Everything on the face comes from them growing toward the midline and fusing. Clefts = places where fusion failed.



The prominences and what each one builds

PROMINENCE	WHERE / WHEN	FORMS
Frontonasal	Between the 1st arches (wk 4–5); mesoderm + NC inside, ectoderm outside	Forehead, bridge of the nose
Medial nasal (x2, merge)	Ridges around the nasal pits (wk 5–8)	Crest + tip of nose, philtrum , intermaxillary segment (primary palate, 4 incisors)
Lateral nasal (x2)	Ridges around the nasal pits (wk 5–8)	Sides (alae) of the nose — NOT the lip
Maxillary (x2)	Lateral to the stomodeum; grow medially and push the nasal prominences to the midline	Lateral upper lip, cheeks (wk 8–9), maxilla , secondary palate (palatine shelves)
Mandibular (x2)	Caudal to the maxillary prominences	Lower lip, chin, lower jaw

Sequence (wk 5–8): nasal placodes (local thickenings beside the frontal prominence) **invaginate** → nasal pits → tissue ridges become medial + lateral nasal prominences → maxillary prominences grow medially → the groove between **medial nasal + maxillary** disappears and they **fuse**. Neural crest forms virtually all the facial bones.

Upper lip & nose

Upper lip = ■ 2 medial nasal (philtrum) + ■ 2 maxillary (lateral lip)

Nose = 5 prominences: ■ frontonasal (bridge) + ■ 2 merged medial nasal (crest, tip) + ■ 2 lateral nasal (sides)

Intermaxillary segment (deep fusion; continuous with the nasal septum), 3 parts:

- **Labial** → philtrum
- **Upper jaw** → 4 incisor teeth
- **Palatal** → triangular **primary palate**

Palate

	PRIMARY	SECONDARY (DEFINITIVE)
From	Intermaxillary segment (medial nasal)	Palatine shelves = inner outgrowths of the maxillary prominences
When	Wk 5–8	Shelves appear wk 6.5–7
How	Triangle in front of the incisive foramen; carries 4 incisors	Grow toward each other above (dorsal to) the tongue and fuse → hard + soft palate , including the uvula at the back end

Nasal cavity, sinuses, teeth

- **Wk 6:** nasal pits deepen into NC and mesoderm
- **Wk 7:** **oronasal membrane** separates nose from mouth → ruptures mid-wk 7 → **primitive choana** (behind the primary palate)
- **Paranasal sinuses:** diverticula of the lateral nasal wall into maxilla, ethmoid, frontal, sphenoid; condition air; grow through puberty and shape the face
- **Teeth** (not tested in class; Step 1 FYI): wk 6 **dental lamina** → 10 buds/jaw → cap stage; milk teeth first; permanent buds wk 8–15

Facial clefts — which fusion failed?

DEFECT	FAILED FUSION	NOTES
Lateral cleft lip	Maxillary + medial nasal	~1/1,000 births, 80% male ; anterior deformity
Cleft secondary palate + cleft uvula	Palatine shelves with each other	Posterior deformity
Oblique facial cleft	Maxillary + lateral nasal	Cleft lip appears here even though the lateral nasal doesn't form lip (secondary to the cleft palate)
Median cleft of upper lip	The two medial nasal prominences	Also: median lower-lip cleft, bifid nose
Macrostomia / microstomia	Maxillary + mandibular (mouth corners)	Macrostomia = NC migration defect

All facial clefts can be caused by improper neural crest migration / function **EXAM EMPHASIS** · Facial clefts are a guaranteed exam topic; detectable on **prenatal imaging** with a skilled technician · **Treatment:** surgery is the only option to date.

Drugs linked to oral clefts

- **Lamotrigine:** 2.5 vs 0.37 per 1,000 (registry data)
 - **Phenobarbital** (older studies)
 - **Valproic acid, lithium:** ↑ craniofacial disorders
- Women on anticonvulsants should be counseled before pregnancy.

Also in this lecture

- **Treacher Collins: Tcof-1** → NC from R2/R4 fails to migrate (~2–3.5 wk)
- **Down syndrome on ultrasound:** **absent nasal bone**, abnormal ear position

Hook: “Lip = Medial + Maxillary (the two M’s); the Lateral nasal stays out of the Lip.” Cleft lip = the two M’s failed to meet. Cleft palate = the palatine shelves failed to meet over the tongue.

★ **USMLE high-yield:** Cleft lip = failed fusion of **maxillary + medial nasal** (primary palate). Cleft palate = failed fusion of **palatine shelves** with each other / nasal septum / primary palate. Separate embryologic causes, but often occur together. Teratogen risk: **phenytoin, valproate, isotretinoin**. Cleft palate → feeding problems, recurrent otitis media (eustachian tube dysfunction).